

SUPPLY CHAIN ANALYSIS REPORT

Project Title

Supply Chain Analysis using Python, SQL, and Power BI

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1. Problem Statement

Supply chain companies generate a large amount of data every day, including customer information, product details, sales, orders, and shipping records. Analyzing this data manually is difficult and time-consuming. The purpose of this project is to clean, analyze, and visualize the supply chain data to identify sales trends, profitable products, customer behavior, and delivery performance. These insights can help businesses make better decisions and improve their overall performance.

2. Objectives

The objectives of this project are:

- To clean and prepare the dataset for analysis.
 - To analyze sales and profit performance.
 - To identify the top-selling products and categories.
 - To compare sales across different markets.
 - To study delivery performance and late delivery risk.
 - To create an interactive dashboard using Power BI.
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3. Dataset Description

The dataset contains **180,519 rows** and **53 columns**.

The dataset includes information such as:

- Customer Details
- Product Details

- Sales
 - Profit
 - Orders
 - Shipping Information
 - Markets
 - Categories
 - Delivery Status
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4. Tools and Technologies Used

- Python
 - Pandas
 - Matplotlib
 - PostgreSQL
 - Power BI
 - Jupyter Notebook
 - GitHub
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5. Methodology

The project was completed in the following steps:

Step 1: Data Cleaning

- Created a copy of the dataset.
- Removed columns with many missing values.
- Converted date columns into DateTime format.
- Checked for missing values and duplicate records.
- Renamed columns for easier analysis.

Step 2: Exploratory Data Analysis (EDA)

Performed analysis to find:

- Total Sales
- Total Profit
- Total Customers
- Total Orders
- Sales by Category
- Sales by Market
- Monthly Sales Trend

- Top Products by Sales
- Profit by Category
- Late Delivery Risk

Step 3: SQL Analysis

Used PostgreSQL to answer business questions by writing SQL queries.

Step 4: Dashboard Creation

Built an interactive Power BI dashboard with KPI cards, charts, and slicers.

6. Results and Insights

Some important findings from the project are:

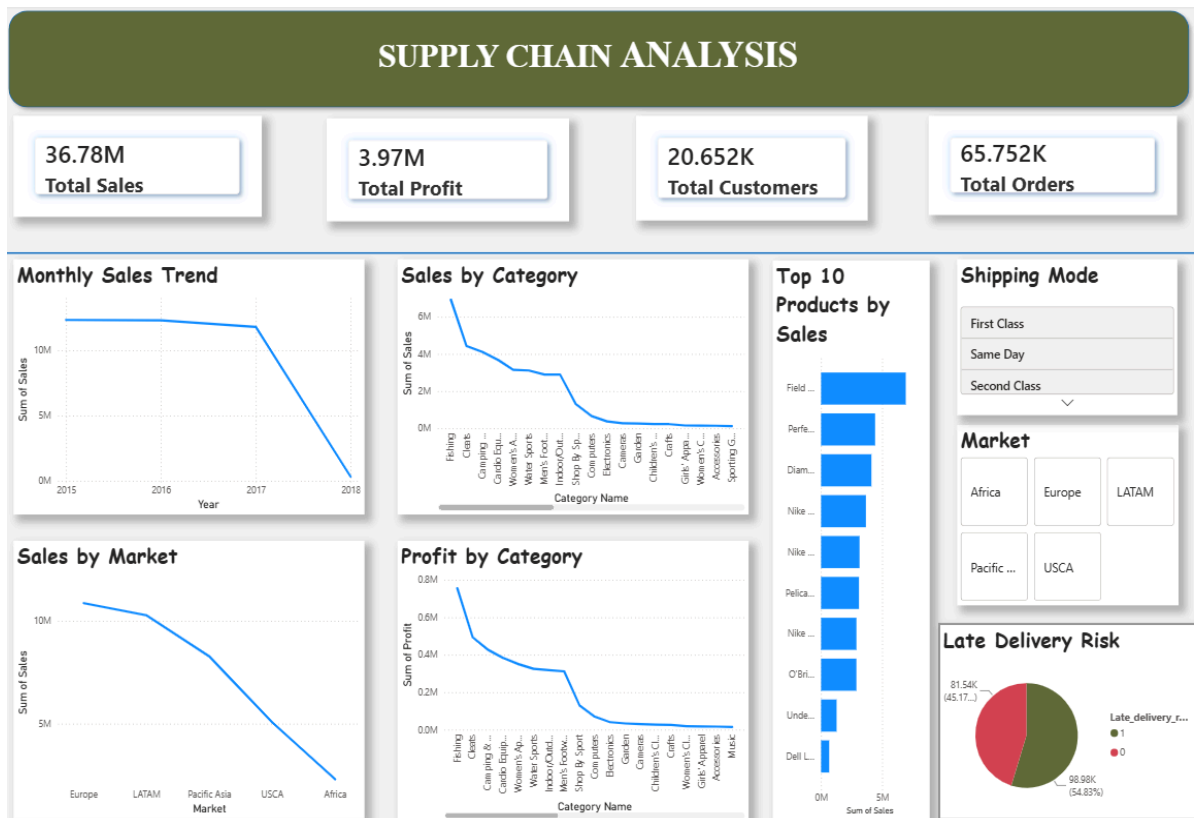
- Europe recorded the highest sales.
 - Fishing was the highest-selling product category.
 - A few products generated a large share of total sales.
 - Sales varied across different years.
 - Many orders had a late delivery risk.
 - Shipping mode affected delivery performance.
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7. Dashboard

The Power BI dashboard includes:

- Total Sales
- Total Profit
- Total Orders
- Total Customers
- Monthly Sales Trend
- Sales by Category
- Sales by Market
- Profit by Category
- Top 10 Products by Sales
- Late Delivery Risk

- Market and Shipping Mode slicers



8. Conclusion

This project helped me understand the complete data analytics process, from data cleaning and analysis to visualization. By using Python, SQL, and Power BI, I gained practical experience in working with real-world data and presenting business insights through an interactive dashboard.

9. Future Scope

- Build a sales forecasting model.
- Use machine learning to predict late deliveries.
- Add more interactive dashboard features using DAX.
- Deploy the dashboard using Power BI Service.